İZMİR DEMOKRASİ UNIVERSITY



MIS203 / WEEK 06

Lists, Tuples And Sets

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A List Stores A Sequence

Order matters when positions matter.

1. Index 0 is the first item.
2. append() adds an item at the end.
3. A for loop visits every item.

Example: examples/01_list_basics.py

Collect And Summarize Sales

A list keeps all values for later questions.

```
sales = [120, 95, 180]
print(sales[0])
print(len(sales))
print(sum(sales))
for sale in sales:
    print(sale)
```

Add a fourth sale and predict the new total.

A Tuple Represents One Fixed Record

Its contents are not changed in place.

1. Keep related values together.
2. Unpack them into named variables.
3. Use a list when records will be added or removed.

Example: examples/03_tuple_records.py

A Product Record

Unpacking makes each field clear.

```
product = ("P-101", "Notebook", 75.0)
code, name, price = product
print(f"{code}: {name}")
print(f"Price: {price:.2f} TRY")
```

Create a second product tuple.

A Set Keeps Unique Values

Membership is the useful question here.

1. Duplicates disappear in a set.
2. Use in to check whether a value exists.
3. Do not rely on set order for a report.

Example: examples/04_set_uniqueness.py

Detect A Duplicate Registration

Normalize the email before checking it.

```
emails = {"ada@example.com"}
new_email = input("Email: ").strip().lower()
if new_email in emails:
    print("Already registered")
else:
    emails.add(new_email)
```

Try the same address with capital letters.

List Or Set?

Pick the collection from the job.

```
visits = ["C01", "C02", "C01"]
unique = set(visits)
print(len(visits))
print(len(unique))
```

Why are the two counts different?

Project A: Sales List Report

Keep sale values and inspect the full series.

projects/project_a_sales_list_report.py

Explain why a running total alone is not enough.

Project B: Event Registration Desk

Store records and reject duplicate email addresses.

projects/project_b_event_registration.py

Test an empty event and a duplicate.

In-Class Lab

Your Challenge

Assignment: week06/lab-quiz/lab_assignment.md

Submit in your repository: week06/lab-quiz/

GitHub Checkpoint

Lab and quiz: week06/lab-quiz/

Keep weekly homework in week06/ outside that folder.

```
git add week06/lab-quiz/ README.md
git commit -m "week06: complete lab quiz"
git push
```

Show the lab, one test and the commit.

See You Next Week